

	hence, rate of increase of kinetic energy = $1200000 - 10000(9.0) = 1.1 \text{ MW}$
8	B $T = 15.9 \text{ days} = 15.9 \times 24 \times 3600 \text{ s}$ $r = 1.22 \times 10^9 \text{ m}$ centripetal acceleration = $r\omega^2$ $= (1.22 \times 10^9)(2\pi/(15.9 \times 24 \times 3600))^2$ $= 2.55 \times 10^{-2} \text{ m s}^{-2}$
9	A